2.2.2 Absence of True Dynamic Pricing

Quotes are static and inflexible. They do not consider **real-time factors** such as traffic congestion, fuel prices, demand surges, or seasonal variations. This results in either inflated costs or unprofitable deals for providers.

2.2.3 Limited AI-Powered Support

Although some platforms deploy basic **lead-generation chatbots**, these are incapable of handling **complex post-booking support**, such as tracking orders, resolving claims, or offering personalized guidance.

2.2.4 Fragmented User Experience

Currently, customers must **switch between multiple channels** (apps, emails, and phone calls) to manage a single move. A **single unified platform** that manages everything—from inventory creation to tracking and support—remains absent.

These gaps collectively indicate a strong need for a **comprehensive, AI-driven solution** like **Shifty**.

2.3. Theoretical Background: Machine Learning in Logistics

Machine Learning (ML) plays a critical role in modern logistics by enhancing decision-making, predicting outcomes, and optimizing resources. For **Shifty**, three main ML applications are identified: **classification, regression, and natural language processing (NLP)**.

2.3.1 Classification Algorithms for Recommendation

The task of recommending the best service package is a **multi-class classification problem**. The input consists of features such as total volume, number of fragile items,